

module3_lab3

September 20, 2026

```
[1]: import pandas as pd
```

```
[2]: df=pd.read_csv("Downloads/bridges.csv")
```

```
[3]: #Assumptions
      #bridge data is correct and can be verified at a later time as they are
      ↳physical structures if question arise
      #will use std from df.describe to see how much of the data is within range
```

```
[4]: df.head()
```

```
[4]:
```

	Id	river	location	erected	purpose	length	lanes	clear-g	t-or-d	\
0	E1	M	3.0	1818	HIGHWAY	NaN	2.0	N	THROUGH	
1	E2	A	25.0	1819	HIGHWAY	1037.0	2.0	N	THROUGH	
2	E3	A	39.0	1829	AQUEDUCT	NaN	1.0	N	THROUGH	
3	E5	A	29.0	1837	HIGHWAY	1000.0	2.0	N	THROUGH	
4	E6	M	23.0	1838	HIGHWAY	NaN	2.0	N	THROUGH	

	material	span	rel-l	type
0	WOOD	SHORT	S	WOOD
1	WOOD	SHORT	S	WOOD
2	WOOD	NaN	S	WOOD
3	WOOD	SHORT	S	WOOD
4	WOOD	NaN	S	WOOD

```
[5]: #13 columns will check on shape to verify
```

```
[6]: df.shape
```

```
[6]: (108, 13)
```

```
[7]: #has 108 rows with 13 columns
```

```
[8]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 108 entries, 0 to 107
Data columns (total 13 columns):
#   Column      Non-Null Count  Dtype
#
```

```

---  -----  -----  -----
0  Id      108 non-null  str
1  river   108 non-null  str
2  location 107 non-null  float64
3  erected 108 non-null  int64
4  purpose 108 non-null  str
5  length  81 non-null  float64
6  lanes   92 non-null  float64
7  clear-g 106 non-null  str
8  t-or-d  102 non-null  str
9  material 106 non-null  str
10 span    92 non-null  str
11 rel-l   103 non-null  str
12 type    106 non-null  str
dtypes: float64(3), int64(1), str(9)
memory usage: 14.7 KB

```

```
[9]: #there are missing values at location (1), length (27),lanes (16), clear g (2),
      ↪ t or d (6),material (2), span (16), rel (5), and type (2)
```

```
[10]: df.describe()
```

```
[10]:
```

	location	erected	length	lanes
count	107.000000	108.000000	81.000000	92.000000
mean	25.978505	1905.314815	1567.469136	2.630435
std	13.665871	37.173664	747.491523	1.164657
min	1.000000	1818.000000	804.000000	1.000000
25%	15.500000	1884.000000	1000.000000	2.000000
50%	27.000000	1903.000000	1300.000000	2.000000
75%	37.500000	1928.000000	2000.000000	4.000000
max	52.000000	1986.000000	4558.000000	6.000000

```
[11]: #find out if there are extreme outliers
```

```
[12]: long_length = df[df['length'] > 2000]
```

```
[13]: print(long_length)
```

	Id	river	location	erected	purpose	length	lanes	clear-g	t-or-d	\
31	E34	O	41.0	1888	RR	4558.0	2.0	G	THROUGH	
41	E42	M	9.0	1895	HIGHWAY	2367.0	2.0	G	THROUGH	
44	E46	A	37.0	1897	RR	4000.0	2.0	G	DECK	
45	E45	M	14.0	1897	RR	2264.0	NaN	G	THROUGH	
52	E87	A	35.0	1903	RR	3000.0	2.0	G	THROUGH	
64	E92	M	10.0	1914	RR	2210.0	NaN	G	THROUGH	
65	E61	O	41.0	1915	RR	2822.0	2.0	G	THROUGH	
67	E62	A	37.0	1918	RR	2300.0	2.0	N	DECK	
68	E63	A	31.0	1920	RR	2122.0	2.0	G	THROUGH	

71	E66	A	32.0	1924	HIGHWAY	2365.0	4.0	G	THROUGH
76	E72	M	5.0	1927	HIGHWAY	2663.0	4.0	N	DECK
78	E75	A	30.0	1928	HIGHWAY	2678.0	4.0	G	DECK
79	E74	M	20.0	1928	HIGHWAY	2220.0	2.0	G	DECK
81	E68	M	17.0	1928	HIGHWAY	2250.0	2.0	G	THROUGH
90	E103	O	48.0	1945	HIGHWAY	2160.0	2.0	G	THROUGH
95	E81	M	14.0	1951	HIGHWAY	2423.0	4.0	G	DECK
97	E88	A	37.0	1955	HIGHWAY	2300.0	4.0	N	DECK
102	E85	M	9.0	1962	HIGHWAY	2213.0	4.0	G	DECK
104	E91	O	44.0	1975	HIGHWAY	3756.0	6.0	G	THROUGH

	material	span	rel-l	type
31	STEEL	LONG	F	SIMPLE-T
41	STEEL	LONG	F	SIMPLE-T
44	STEEL	LONG	F	SIMPLE-T
45	STEEL	NaN	F	SIMPLE-T
52	STEEL	MEDIUM	S	SIMPLE-T
64	STEEL	MEDIUM	F	SIMPLE-T
65	STEEL	LONG	F	SIMPLE-T
67	STEEL	LONG	F	CONT-T
68	STEEL	MEDIUM	F	SIMPLE-T
71	STEEL	MEDIUM	S	ARCH
76	STEEL	MEDIUM	S-F	CANTILEV
78	STEEL	MEDIUM	F	ARCH
79	STEEL	MEDIUM	S-F	CANTILEV
81	STEEL	MEDIUM	S	SIMPLE-T
90	STEEL	LONG	F	CANTILEV
95	STEEL	LONG	F	CONT-T
97	STEEL	LONG	F	CONT-T
102	STEEL	LONG	F	CONT-T
104	STEEL	LONG	F	ARCH

```
[14]: #trying to see how extreme the values are above the 75% for std and can assume
      ↪ these longer briges are related to railroad (RR) and wouldn't be consider
      ↪ extreme. But possible assumption would be outside of this range could prove
      ↪ to be an outlier
```

```
[15]: df.isnull().sum()
```

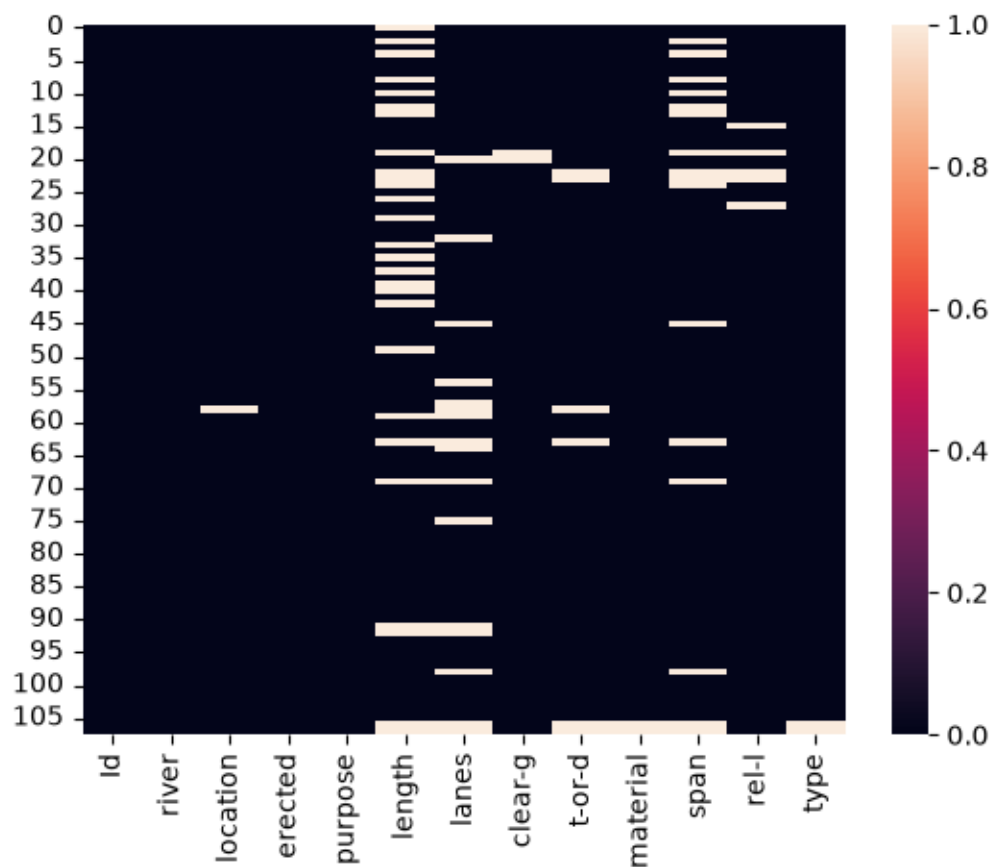
```
[15]: Id          0
      river      0
      location   1
      erected    0
      purpose    0
      length     27
      lanes      16
      clear-g    2
```

```
t-or-d      6
material    2
span       16
rel-l       5
type        2
dtype: int64
```

```
[16]: import seaborn as sns
```

```
[17]: sns.heatmap(df.isnull())
```

```
[17]: <Axes: >
```



```
[18]: print(df[df['span']=='SHORT'][['Id','span','length']])
```

```
   Id  span  length
0  E1 SHORT    NaN
1  E2 SHORT 1037.0
3  E5 SHORT 1000.0
6  E8 SHORT 1000.0
```

```

7    E9  SHORT  1500.0
16   E18 SHORT  1200.0
21   E22 SHORT  1200.0
33   E36 SHORT    NaN
38   E4  SHORT  1092.0

```

```
[19]: short_bridge=df[df['span']=='SHORT']
```

```
[20]: print(short_bridge.describe())
```

	location	erected	length	lanes
count	9.000000	9.000000	7.000000	9.000000
mean	23.555556	1854.000000	1147.000000	2.000000
std	13.172994	28.057976	177.280381	0.866025
min	3.000000	1818.000000	1000.000000	1.000000
25%	24.000000	1837.000000	1018.500000	2.000000
50%	27.000000	1846.000000	1092.000000	2.000000
75%	28.000000	1876.000000	1200.000000	2.000000
max	45.000000	1892.000000	1500.000000	4.000000

```
[21]: #need to check for what are short bridges with length min - 1000 max - 1500
```

```
[22]: print(df[df['span']=='MEDIUM'][['Id','span','length']])
```

	Id	span	length
5	E7	MEDIUM	990.0
9	E11	MEDIUM	1000.0
11	E14	MEDIUM	1200.0
14	E16	MEDIUM	1030.0
15	E17	MEDIUM	1000.0
17	E19	MEDIUM	1000.0
18	E20	MEDIUM	1000.0
25	E26	MEDIUM	1150.0
26	E30	MEDIUM	NaN
27	E29	MEDIUM	1080.0
28	E28	MEDIUM	1000.0
29	E32	MEDIUM	NaN
30	E31	MEDIUM	1161.0
32	E33	MEDIUM	1120.0
34	E35	MEDIUM	1000.0
35	E38	MEDIUM	NaN
36	E37	MEDIUM	1350.0
37	E39	MEDIUM	NaN
39	E40	MEDIUM	NaN
40	E41	MEDIUM	NaN
46	E47	MEDIUM	2000.0
47	E58	MEDIUM	1200.0
48	E48	MEDIUM	2000.0
50	E49	MEDIUM	1850.0

```

51      E95  MEDIUM  1300.0
52      E87  MEDIUM  3000.0
53      E51  MEDIUM  1417.0
55      E89  MEDIUM  1200.0
56      E53  MEDIUM   965.0
58      E54  MEDIUM  1240.0
59      E56  MEDIUM    NaN
64      E92  MEDIUM  2210.0
68      E63  MEDIUM  2122.0
70      E64  MEDIUM   885.0
71      E66  MEDIUM  2365.0
72      E70  MEDIUM   860.0
73      E69  MEDIUM   884.0
75      E73  MEDIUM  1508.0
76      E72  MEDIUM  2663.0
78      E75  MEDIUM  2678.0
79      E74  MEDIUM  2220.0
80      E71  MEDIUM   860.0
81      E68  MEDIUM  2250.0
86      E79  MEDIUM  1800.0
87      E108 MEDIUM  1060.0
88      E107N MEDIUM   840.0
89      E105 MEDIUM  1710.0
91      E97  MEDIUM    NaN
92      E96  MEDIUM    NaN
93      E99  MEDIUM  1320.0
94      E98  MEDIUM   900.0
101     E86  MEDIUM   980.0
103     E84  MEDIUM   870.0

```

```
[23]: medium_bridge=df[df['span']=='MEDIUM']
      print(medium_bridge.describe())
```

	location	erected	length	lanes
count	52.000000	53.000000	44.000000	46.000000
mean	24.609615	1907.207547	1414.500000	2.608696
std	11.578941	29.516852	583.565444	1.021508
min	3.000000	1840.000000	840.000000	2.000000
25%	17.000000	1889.000000	1000.000000	2.000000
50%	26.000000	1903.000000	1180.500000	2.000000
75%	32.000000	1928.000000	1812.500000	4.000000
max	52.000000	1969.000000	3000.000000	6.000000

```
[24]: #some "impossible" data where a medium bridge is min - 840 and max is 3000
```

```
[25]: print(df[df['span']=='LONG'][['Id','span','length']])
```

	Id	span	length
20	E23	LONG	1245.0

```

31    E34    LONG    4558.0
41    E42    LONG    2367.0
42    E44    LONG         NaN
43    E43    LONG    1040.0
44    E46    LONG    4000.0
49    E94    LONG         NaN
54    E50    LONG    1154.0
57    E52    LONG    1504.0
60    E55    LONG    1730.0
61    E57    LONG    1620.0
62    E59    LONG    1652.0
65    E61    LONG    2822.0
66    E60    LONG    1000.0
67    E62    LONG    2300.0
74   E101    LONG    1770.0
77    E67    LONG    1330.0
82    E78    LONG    1365.0
83    E77    LONG    1450.0
84    E76    LONG    1500.0
85    E93    LONG    1690.0
90   E103    LONG    2160.0
95    E81    LONG    2423.0
96    E80    LONG    1031.0
97    E88    LONG    2300.0
99   E102    LONG    1700.0
100   E83    LONG    1000.0
102   E85    LONG    2213.0
104   E91    LONG    3756.0
105   E90    LONG     950.0

```

```

[26]: long_bridge=df[df['span']=='LONG']
      print(long_bridge.describe())

```

	location	erected	length	lanes
count	30.000000	30.00000	28.000000	27.000000
mean	26.033333	1925.10000	1915.357143	3.185185
std	17.697620	27.20592	925.831247	1.387777
min	1.000000	1876.00000	950.000000	2.000000
25%	9.000000	1903.25000	1308.750000	2.000000
50%	30.000000	1922.50000	1671.000000	2.000000
75%	41.750000	1949.50000	2300.000000	4.000000
max	49.000000	1978.00000	4558.000000	6.000000

```

[27]: # longest bridge at min is 950 and max is at 4558, there seems to be a crossing
      ↪ of c=boundaries between the medium and long bridges

```

```

[28]: missing_length = df[df['length'].isnull()]

```

```
[29]: print(missing_length)
```

	Id	river	location	erected	purpose	length	lanes	clear-g	t-or-d	\
0	E1	M	3.0	1818	HIGHWAY	NaN	2.0	N	THROUGH	
2	E3	A	39.0	1829	AQUEDUCT	NaN	1.0	N	THROUGH	
4	E6	M	23.0	1838	HIGHWAY	NaN	2.0	N	THROUGH	
8	E10	A	39.0	1848	AQUEDUCT	NaN	1.0	N	DECK	
10	E12	A	39.0	1853	RR	NaN	2.0	N	DECK	
12	E13	A	33.0	1856	HIGHWAY	NaN	2.0	N	THROUGH	
13	E15	A	28.0	1857	RR	NaN	2.0	N	THROUGH	
19	E21	M	16.0	1874	RR	NaN	2.0	NaN	THROUGH	
22	E24	O	45.0	1878	RR	NaN	2.0	G	NaN	
23	E25	M	10.0	1882	RR	NaN	2.0	G	NaN	
24	E27	A	39.0	1883	RR	NaN	2.0	G	THROUGH	
26	E30	A	31.0	1884	RR	NaN	2.0	G	THROUGH	
29	E32	A	30.0	1887	HIGHWAY	NaN	2.0	G	THROUGH	
33	E36	O	45.0	1890	HIGHWAY	NaN	2.0	G	THROUGH	
35	E38	M	17.0	1891	HIGHWAY	NaN	2.0	G	THROUGH	
37	E39	A	25.0	1892	HIGHWAY	NaN	2.0	G	THROUGH	
39	E40	M	22.0	1893	HIGHWAY	NaN	2.0	G	THROUGH	
40	E41	M	11.0	1894	HIGHWAY	NaN	2.0	G	THROUGH	
42	E44	O	48.0	1896	HIGHWAY	NaN	2.0	G	THROUGH	
49	E94	M	13.0	1901	RR	NaN	2.0	G	THROUGH	
59	E56	M	23.0	1909	HIGHWAY	NaN	NaN	G	THROUGH	
63	E107	A	39.0	1914	RR	NaN	NaN	G	NaN	
69	E65	A	30.0	1921	WALK	NaN	NaN	G	THROUGH	
91	E97	Y	52.0	1945	HIGHWAY	NaN	NaN	G	THROUGH	
92	E96	Y	51.0	1945	RR	NaN	NaN	G	THROUGH	
106	E100	O	43.0	1982	HIGHWAY	NaN	NaN	G	NaN	
107	E109	A	28.0	1986	HIGHWAY	NaN	NaN	G	NaN	

	material	span	rel-l	type
0	WOOD	SHORT	S	WOOD
2	WOOD	NaN	S	WOOD
4	WOOD	NaN	S	WOOD
8	WOOD	NaN	S	WOOD
10	WOOD	NaN	S	WOOD
12	WOOD	NaN	S	WOOD
13	WOOD	NaN	S	WOOD
19	IRON	NaN	NaN	SIMPLE-T
22	STEEL	NaN	NaN	SIMPLE-T
23	STEEL	NaN	NaN	SIMPLE-T
24	STEEL	NaN	F	SIMPLE-T
26	STEEL	MEDIUM	F	SIMPLE-T
29	IRON	MEDIUM	F	SIMPLE-T
33	IRON	SHORT	F	SIMPLE-T
35	IRON	MEDIUM	F	SIMPLE-T
37	STEEL	MEDIUM	F	SIMPLE-T

39	STEEL	MEDIUM	F	SIMPLE-T
40	IRON	MEDIUM	F	SIMPLE-T
42	STEEL	LONG	F	SUSPEN
49	STEEL	LONG	F	SIMPLE-T
59	STEEL	MEDIUM	F	SIMPLE-T
63	STEEL	NaN	F	NIL
69	STEEL	NaN	F	SUSPEN
91	STEEL	MEDIUM	S	ARCH
92	STEEL	MEDIUM	F	SIMPLE-T
106	NaN	NaN	F	NaN
107	NaN	NaN	F	NaN

[30]: *#these are the isolated values of missing legth that would need to be*
→investigated than filling out since 27 out of 108 values is almost 25% of
→the information is missing. Filling in values will skew data be it median
→or mean will not work
Will need to investigate or followup on bridge length as this will introduce
→bias if not properly looked at. Will keep current values until more
→information is present